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Analogue and Digital Communication

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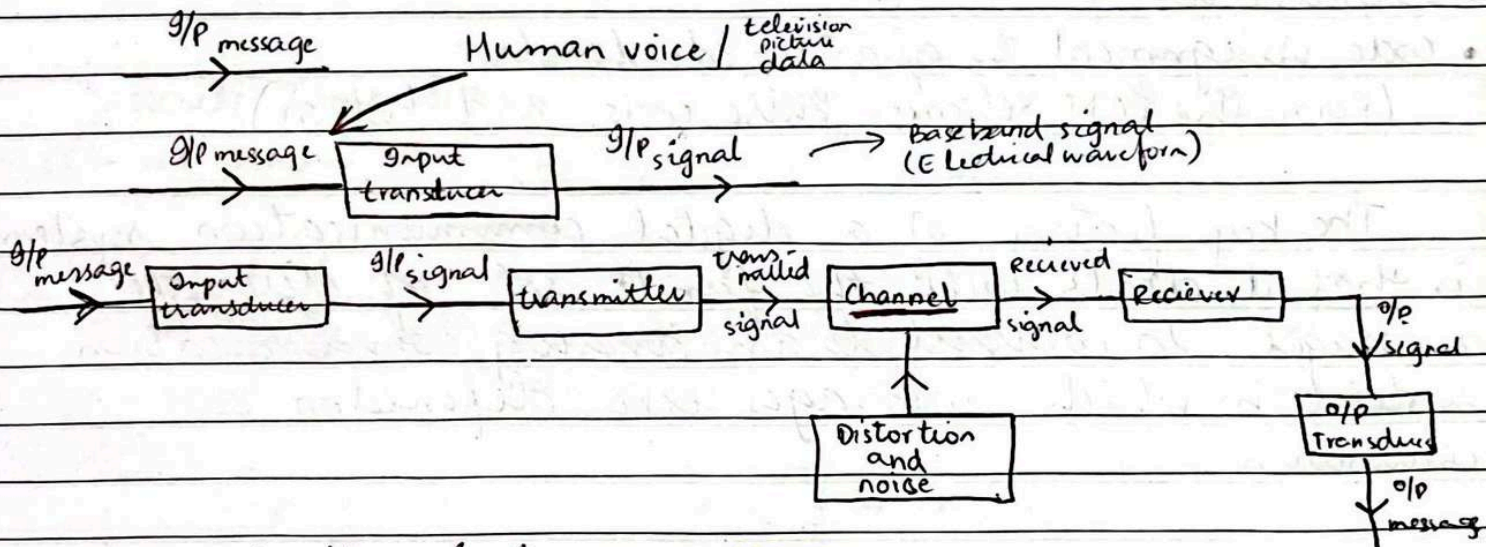
DAY-1

Book:- Modern Digital and analogue Communication systems, 3rd Edition, B. P. Lathi

Ch # 1 - Ch # 6

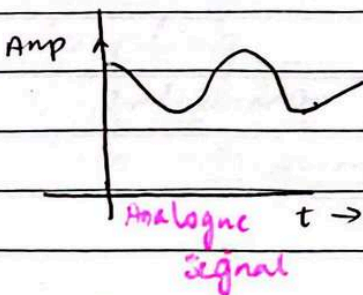
Note:- Chapter 1 is important

Chapter 1:-



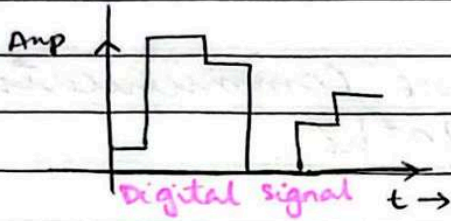
Communication System S-YAO

SNR (Signal To Noise Ratio) is the ratio of signal power to noise power. Noise accumulates along the path. Noise level increases with distance from the transmitter.



Message defined on a continuum (e.g. microphone)

DAY-1



Finite set of possible messages
 (e.g. typewriter)

Analogue - to - digital Conversion :-

It consists of these steps.

- Sampling
- Quantization
- Code Assignment to quantized levels
 (Form the PCM Scheme - Pulse code modulation)

The key feature of a digital communication system is that it deals with the finite set of discrete messages. In contrast to an analog communication system in which messages are defined on a continuum.

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1. Analog Signal :-

An analog signal is any continuous-time and continuous amplitude signal.

$$s(t) \quad t \in \mathbb{R}$$

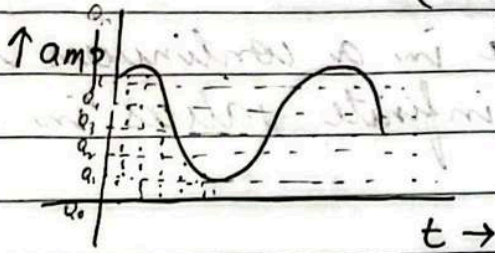
For mathematical treatment

$$s \in \mathbb{C}$$

2. Discrete Time signal :-

These are signals defined on a discrete time domain $\{nT_s \mid n \in \mathbb{Z}\}$ and assume continuous values on amplitude.

$$s(nT_s) \in \mathbb{C}, n \in \mathbb{Z}$$



Discrete time signal are derived by sampling at instants that are multiple of T_s .

3. Digital signal :-

A digital signal is a discrete signal in both time domain and amplitude, so a digital signal satisfies.

$$s(nT_s) \in A, n \in \mathbb{Z}, A = \{Q_0, Q_1, Q_2, \dots, Q_n\} \text{ with } A \text{ is a set of } n \text{ elements!}$$

Analog-to-Digital conversion of signals :-

Conversion of signals from analog-to-digital domain requires

- 1) Sampling
- 2) Quantization
- 3) Code assignment

→ The sampling theorem states that "if the highest frequency in the signal spectrum is B , (in Hertz), the signal can be reconstructed from its samples taken at a rate of $2B$ samples per second."

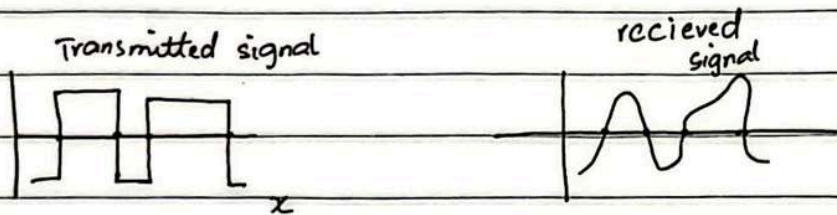
→ The sample value is still lie in a continuous range and can take anyone of the infinite values in the range.

→ The problem is resolved by quantization where everyone of the samples is rounded off to the nearest quantized level.

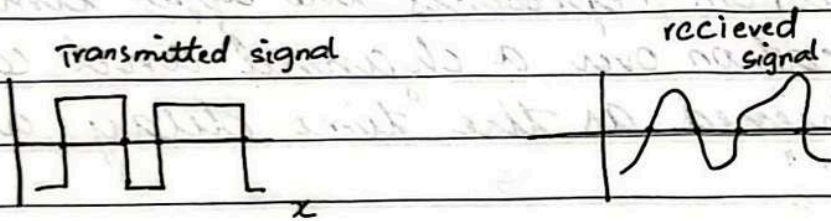
→ If 16 levels are assigned, then we could design a code for the transmission of digits.

Digit	Binary Equivalent	Pulse code waveform
0	0 0 0 0	
1	0 0 0 1	
2	0 0 1 0	
⋮		
9	0 0 0 1	
⋮		
15	1 1 1 1	

- The scheme of transmitting data by digitizing and then using pulse codes to transmit the digitized data is known as **pulse code modulation (PCM)**
- Major advantage of digital communication is the possibility of "regeneration of signals."



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- Major advantage of digital communication is the possibility of "regeneration of signals."



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Shannon's Theorem:-

$$C = B \log_2 (1 + \text{SNR})$$

channel capacity

bandwidth

signal-to-noise ratio

- * Rate of information transmission (C) $\propto B$
- * B and SNR are exchangeable.

To maintain a given rate and accuracy of transmission, we trade SNR for B but not vice versa.

$$\text{SNR}_2 \approx \text{SNR}_1 \frac{B_1}{B_2}$$

Example:-

$$\text{SNR}_1 = 64 \quad B_1 = 4 \Rightarrow B_2 = 8 \quad \text{SNR} = 8$$

(small increase in B buys a large advantage in SNR)

Rate of information transmission or channel capacity C

in simplest words gives the maximum number of bits that can be transmitted per second with $P_e \rightarrow 0$

If $N = 0$, then $C \rightarrow \infty$

Shannon's theorem represents the upper limit on rate of communication over a channel and cannot be practically achieved as the "time delay and complexity $\rightarrow \infty$ "

* Two primary communication sources are bandwidth (B) and the transmitted power (SNR)

Significance of C:-

If the information rate $R < C$ then it is theoretically possible to achieve reliable transmission [$P_e \rightarrow 0$] through the channel by appropriate channel coding
 If $R > C$, reliable transmission is not possible

END OF CHAPTER 1

Chapter #2 :-

Signal Energy and Energy Spectral Density:-

$$E_g = \int_{-\infty}^{\infty} g^2(t) dt$$

For complex signals $E_g = \int_{-\infty}^{\infty} |g(t)|^2 dt$

The energy E_g of a signal $g(t)$ is defined as the area under $|g(t)|^2$

Signal energy is also determined from the Fourier Transform $G(\omega)$ through the Parseval's theorem.

$$g(t) \leftrightarrow G(\omega)$$
$$G(\omega) = \int_{-\infty}^{\infty} g(t) e^{-j\omega t} dt$$

$$g(t) = \frac{1}{2\pi} \int_{-\infty}^{\infty} G(\omega) e^{j\omega t} d\omega$$

$$E_g = \int_{-\infty}^{\infty} g(t) g^*(t) dt = \int_{-\infty}^{\infty} g(t) \left[\frac{1}{2\pi} \int_{-\infty}^{\infty} G(\omega) e^{j\omega t} d\omega \right] dt$$

$g^*(t)$ being the conjugate of $g(t)$ can be expressed by modifying the IFT. Interchanging the order of integration yields

$$E_g = \frac{1}{2\pi} \int_{-\infty}^{\infty} G^*(\omega) \left[\int_{-\infty}^{\infty} g(t) e^{j\omega t} dt \right] d\omega$$

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Signal Energy & Energy Spectral Density (ESD)

$$E_g = \int_{-\infty}^{\infty} |g(t)|^2 dt$$

$$E_g = \frac{1}{2\pi} \int_{-\infty}^{\infty} |G(\omega)|^2 d\omega$$

Energy spectral Density (ESD)

$$\Psi_g(\omega) = |G(\omega)|^2$$

Ex:- Find the signal energy for $g(t) = e^{-at} u(t); a > 0$ by the time domain specification and the equivalent signal spectrum $G(\omega)$.

Solution:-

$$g(t) = e^{-at} u(t)$$

$$G(\omega) = \frac{1}{j\omega + a}$$

$$E_g = \int_{-\infty}^{\infty} g^2(t) dt = \int_0^{\infty} e^{-2at} dt = \frac{1}{2a} \rightarrow \textcircled{A}$$

E_g by the Parseval's theorem

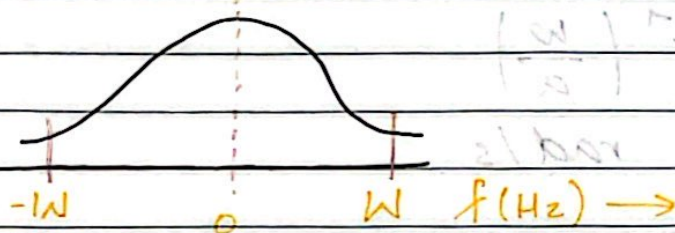
$$\begin{aligned} E_g &= \frac{1}{2\pi} \int_{-\infty}^{\infty} |G(\omega)|^2 d\omega = \frac{1}{2\pi} \int_{-\infty}^{\infty} \frac{1}{\omega^2 + a^2} d\omega \\ &= \frac{1}{2\pi a} \tan^{-1} \frac{\omega}{a} \Big|_{-\infty}^{\infty} = \frac{1}{2a} \rightarrow \textcircled{B} \end{aligned}$$

Hence the parseval's theorem is verified and the signal energy E_g can be found by both time domain $g(t)$ and its fourier transform $G(\omega)$

Essential BW of the signal:-

For most practical signals, the signal spectrum beyond W Hz can be suppressed with little effect on the

signal shape and energy.



Example:-

Estimate the essential bandwidth W rad/s of the signal $e^{-at}u(t)$ if the essential band is required to contain 95% of signal energy.

Solution:-

$$e^{-at}u(t) \longleftrightarrow \frac{1}{j\omega + a}$$

and the ESD $\psi_g(\omega) = |G(\omega)|^2 = \frac{1}{\omega^2 + a^2}$

$$E_g = \frac{1}{2a}$$

$$E_g = \frac{1}{2\pi} \times \text{area under ESD}$$

$$\frac{1}{2a} = \frac{1}{2\pi} \times \text{area under ESD}$$

$$\frac{0.95 \times 1}{2a} = \frac{1}{2\pi} \int_{-W}^W \frac{1}{\omega^2 + a^2} d\omega$$

$$\frac{0.95}{2a} = \frac{1}{2\pi a} \left[\tan^{-1} \frac{\omega}{a} \right]_{-W}^W$$

$$\frac{0.95}{2a} = \frac{1}{\pi a} \tan^{-1} \left(\frac{\omega}{a} \right)$$

$$\frac{0.95\pi}{2} = \tan^{-1} \left(\frac{\omega}{a} \right)$$

$$\omega = 12.706 a \text{ rad/s}$$

→ Spectral components of $g(t)$ from 0 to 12.706 a rad/s (2.02 a Hz) contribute 95% of signal energy.

Signal Power & Power Spectral Density (PSD) :

For power signals [signals with $g(t) \rightarrow 0$ at $t \rightarrow \infty$], the signal power is meaningful as "time average of energy" averaged over an infinite time interval.

The power P_g of a signal $g(t)$ is given by

$$P_g = \lim_{T \rightarrow \infty} \frac{P_{in}}{T} = \frac{1}{T} \int_{-T/2}^{T/2} |g(t)|^2 dt$$

Signal power is thus just the time average of signal energy.

Power Spectral Density (PSD) $S_g(\omega)$:

$S_g(\omega) = \lim_{T \rightarrow \infty} |G_T(\omega)|^2$ and consequently

$$P_g = \frac{1}{2\pi} \int_{-\infty}^{\infty} S_g(\omega) d\omega = \frac{1}{\pi} \int_0^{\infty} S_g(\omega) d\omega$$

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Signal Power & Power Spectral Density (PSD) :-

$$P_g = \lim_{T \rightarrow \infty} \frac{1}{T} \int_{-T/2}^{T/2} |g(t)|^2 dt$$

$$S_g(\omega) = \lim_{T \rightarrow \infty} \frac{|G_T(\omega)|^2}{T}$$

and consequently

$$P_g = \frac{1}{2\pi} \int_{-\infty}^{\infty} S_g(\omega) d\omega$$

The power is $\frac{1}{2\pi}$ times the area under the PSD

PSD is a positive, real and even function of ω .

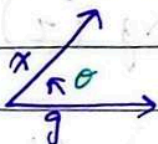
The PSD $S_g(\omega)$ represents the power per unit BW (in Hertz) of the spectral components at frequency ω .

The power contributed by the spectral components within the band ω_1 to ω_2 is given by

$$\Delta P_g = \frac{1}{\pi} \int_{\omega_1}^{\omega_2} S_g(\omega) d\omega$$

Correlation :-

Two vectors g and x are correlated if g has a large component along x .



smaller the θ , larger the correlation

Correlation coefficient

DAY-2

$$C_n = \cos \theta = \frac{g \cdot n}{|g||n|}$$

$$-1 \leq C_n \leq 1$$

For aligned vectors ($C_n = 1$)

For vectors in opposite direction ($C_n = -1$)

For signals $g(t)$ and $n(t)$

$$C_n = \frac{1}{\sqrt{E_g E_n}} \int_{-\infty}^{\infty} g(t) n(t) dt$$

$C_n = 0$ if $g(t)$ and $n(t)$ are orthogonal.

Auto correlation

The correlation of a signal with itself is called "auto correlation"

$$\Psi_g(t) = \int_{-\infty}^{\infty} g(t) g(t-\tau) dt$$

Auto correlation method for a signal $g(t)$, the energy spectral density is $|G(\omega)|^2$ can also be found by taking the fourier transform of its auto correlation function. The reason for this alternate method is to lay a foundation for dealing with random signals.

The time auto correlation function $R_g(\tau)$ of a real power signal $g(t)$ is defined as

$$R_g(\tau) = \lim_{T \rightarrow \infty} \frac{1}{T} \int_{-T/2}^{T/2} g(t) g(t+\tau) dt$$

• $R_g(t)$ is an even function of τ . This means for real $g(t)$

$$R_g(t) = \lim_{T \rightarrow \infty} \frac{1}{T} \int_{-T/2}^{T/2} g(t) g(t-\tau) dt$$

$$R_g(\tau) = R_g(-\tau)$$

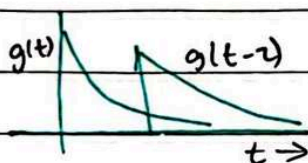
QUESTION:-

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Find the time auto-correlation function of signal $g(t) = e^{-at} u(t)$ and from it determine the ESD of $g(t)$.

Sol:-

$$g(t) = e^{-at} u(t) \text{ and } g(t-\tau) = e^{-a(t-\tau)} u(t-\tau)$$



Auto correlation function $\Psi_g(\tau)$ is given by area under the product $g(t) g(t-\tau)$

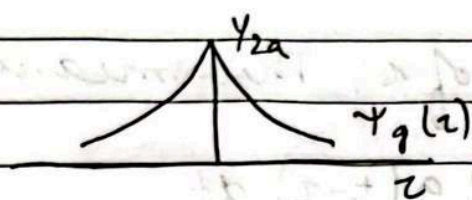
$$\Psi_g(\tau) = \int_{-\infty}^{\infty} g(t) g(t-\tau) dt$$

$$= \int_{-\infty}^{\infty} e^{-at} u(t) e^{-a(t-\tau)} u(t-\tau) dt$$

$$= e^{a\tau} \int_{\tau}^{\infty} e^{-2at} dt = \frac{1}{2a} e^{-a\tau}$$

For real $g(t)$, $\Psi_g(\tau)$ is an even function. so

$$\Psi_g(\tau) = \frac{1}{2a} e^{-a|\tau|}$$



$$ESD \Psi_g(\omega) = \frac{1}{\omega^2 + a^2}$$

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$$P_g = \lim_{T \rightarrow \infty} \frac{1}{T} \int_{-T/2}^{T/2} |g(t)|^2 dt$$

$$S_g(\omega) = \lim_{T \rightarrow \infty} \frac{|G_T(\omega)|^2}{T} \rightarrow \text{PSD}$$

$$P_g = \frac{1}{\pi} \int_0^\infty S_g(\omega) d\omega$$

$$\Delta P_g = \frac{1}{\pi} \int_{\omega_1}^{\omega_2} S_g(\omega) d\omega$$

Correlation:

$$C_n = \cos \theta = \frac{g \cdot n}{|g||n|}$$

$$C_n = \frac{1}{\sqrt{E_g E_n}} \int_{-\infty}^{\infty} g(t) n(t) dt$$

$$-1 \leq C_n \leq 1$$

$$C_n = 1$$

$$C_n = -1$$

$$C_n = 0$$

aligned

opposite

orthogonal

Autocorrelation:-

$$\psi_g(\tau) = \int_{-\infty}^{\infty} g(t) g(t-\tau) dt$$

Energy Spectral Density [ESD] of the input and output of the LTI system :-

If $g(t)$ and $y(t)$ are the input and the corresponding output of a linear time-invariant system (LTI) then

$$y(t) = g(t) * h(t)$$

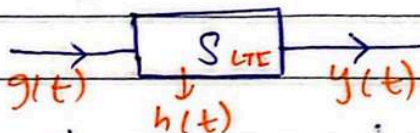
$$Y(\omega) = G(\omega) H(\omega)$$

$$Y(\omega) = H(\omega) G(\omega)$$

$$|Y(\omega)|^2 = |H(\omega)|^2 |G(\omega)|^2$$

This shows that

$$\psi_y(\omega) = |H(\omega)|^2 \psi_g(\omega)$$



This output ESD is $|H(\omega)|^2$ times the input ESD. Since the PSD is a time average of ESD, the input/output signal PSD of an LTI system is

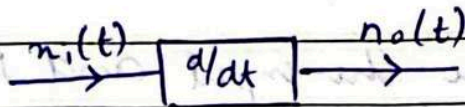
$$S_y(\omega) = |H(\omega)|^2 S_g(\omega)$$

QUESTION:-

A noise signal $n_i(t)$ with PSD $S_{n_i}(\omega) = K$ is

applied at the input of an ideal differentiator. Determine the PSD and the power of output noise signal $n_o(t)$.

Sol.



The transfer function for an ideal differentiator is $H(\omega) = j\omega$

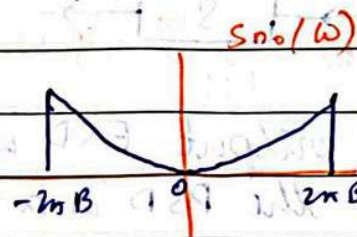
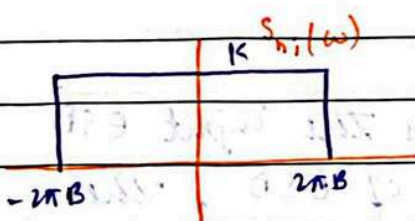
If noise at differentiator output is $n_o(t)$, then

$$S_{n_o}(\omega) = |H(\omega)|^2 S_{n_i}(\omega) = |j\omega|^2 K = \omega^2 K$$

The op PSD is thus parabolic.

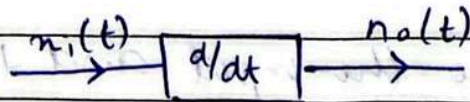
(*) The output noise power N_o is $1/2\pi$ times area under the output PSD.

$$N_o = \frac{1}{2\pi} \int_{-2\pi B}^{2\pi B} K \omega^2 d\omega = \frac{K}{2\pi} \int_{-2\pi B}^{2\pi B} \omega^2 d\omega = \frac{8\pi^2 B^3 K}{3}$$



applied at the input of an ideal differentiator. Determine the PSD and the power of output noise signal $n_o(t)$.

Sol.



The transfer function for an ideal differentiator is $H(\omega) = j\omega$

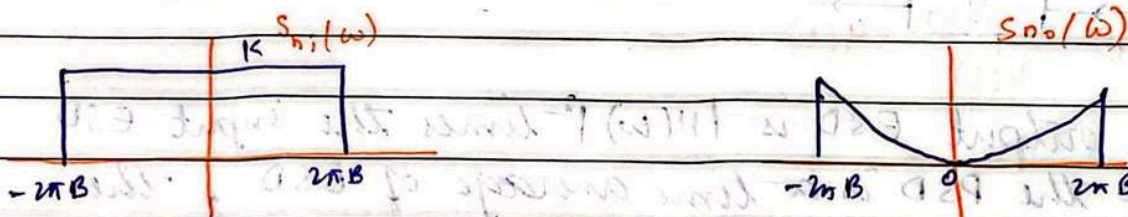
If noise at differentiator output is $n_o(t)$, then

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(*) The output noise power N_o is $1/2\pi$ times area under the output PSD.

$$N_o = \frac{1}{2\pi} \int_{-2\pi B}^{2\pi B} K \omega^2 d\omega = \frac{K}{2\pi} \int_{-2\pi B}^{2\pi B} \omega^2 d\omega = \frac{8\pi^2 B^3 K}{3}$$



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Signal Transmission through a Linear System

For a linear time-invariant (LTI) continuous

time system, the input-output relationship is given by

$$y(t) = g(t) * h(t) \rightarrow \textcircled{A}$$

where $g(t)$ is the input, $y(t)$ is the output and $h(t)$ is the unit-impulse response

$$g(t) \Leftrightarrow G(\omega)$$

$$y(t) \Leftrightarrow Y(\omega)$$

$$h(t) \Leftrightarrow H(\omega)$$

where $H(\omega)$ is the system transfer function, then the application of the time convolution property yields

$$Y(\omega) = G(\omega) H(\omega) \rightarrow \textcircled{B} / (L.S)$$

$H(\omega)$ thus represents the spectral response of the system

$$|Y(\omega)| e^{j\theta_y(\omega)} = |G(\omega)| |H(\omega)| e^{j[\theta_g(\omega) + \theta_h(\omega)]}$$

eq. (L.S) brings out the idea of the spectral shaping (or modification) of the signal by the system

Plots of $|H(\omega)|$ & $\theta_h(\omega)$ as function of ω show how the system modifies the amplitude and the phase respectively

Distortionless Transmission :-

Transmission is distortionless if input and

output have identical waveshapes within a multiplicative constant

Since

$$Y(\omega) = G(\omega) \cdot H(\omega)$$

$$H(\omega) = K e^{-j\omega t_d}$$

$$\rightarrow |H(\omega)| = K \rightarrow \text{constant}$$

$$\theta_K(\omega) = -\omega t_d \rightarrow \text{linear function of } \omega$$

A delayed output that retains the input waveform is also considered distortionless.

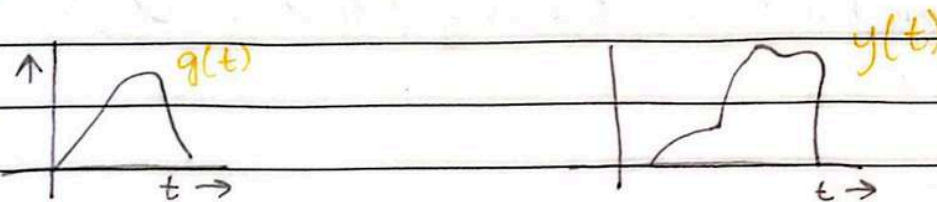
$$Y(t) = K g(t - t_d)$$

For the distortionless case, the output signal is the input signal multiplied by K and delayed by t_d .

For distortionless transmission t_d should be constant over the band of interest.

A system may have flat amplitude response and get hugely distorted signal if the phase response is not linear. (t_d is not constant).

Signal Distortion over a communication Channel:



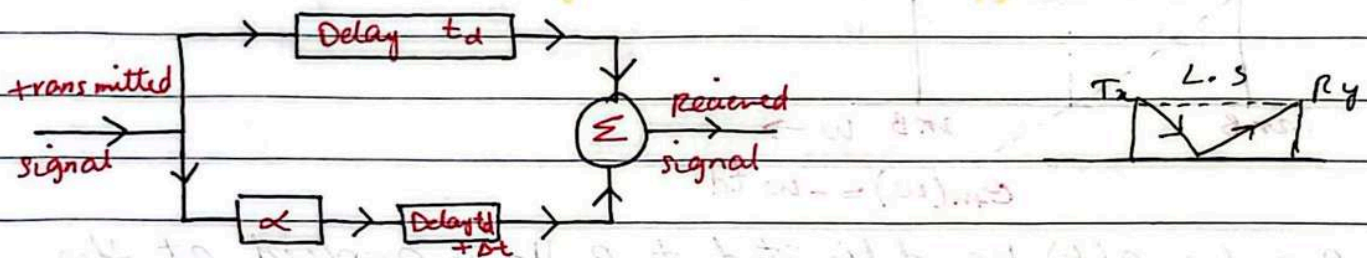
The pulse will spread if the channel characteristics is not ideal that is

$$\phi_c(\omega) \neq -\omega t_d$$

Spreading, or dispersion, of the pulse will occur if either the amplitude response or the phase response or both are non-ideal.

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Distortion Caused By Multipath effect:



Scenario considered has only two paths:

- 1) With a unity gain and delay t_d .
- 2) Other with gain α and delay $(t_d + \Delta t)$

Transfer function of the two paths are given by

$$e^{-j\omega t_d} \text{ and } \alpha \cdot e^{-j\omega(t_d + \Delta t)} \text{ respectively}$$

Overall transfer function $H(\omega)$ is

$$\begin{aligned} H(\omega) &= e^{-j\omega t_d} + \alpha \cdot e^{-j\omega(t_d + \Delta t)} \\ &= e^{-j\omega t_d} (1 + \alpha \cdot e^{-j\omega \Delta t}) \\ &= e^{-j\omega t_d} (1 + \alpha \cos \omega \Delta t - j\alpha \sin \omega \Delta t) \end{aligned}$$

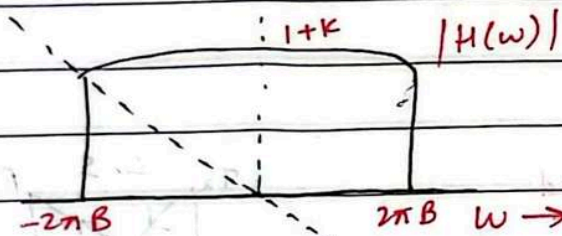
Both magnitude and the phase characteristics

of $H(\omega)$ are periodic in ω with a period of $\frac{2\pi}{\Delta t}$.

Example:

A low pass filter (shown below) has transfer function $H(\omega)$ given by

$$H(\omega) = \begin{cases} (1 + K \cos T\omega) e^{-j\omega t_d} & |\omega| < 2\pi B \\ 0 & |\omega| > 2\pi B \end{cases}$$



$$\phi_m(\omega) = -\omega t_d$$

A pulse $g(t)$ band limited to B Hz is applied at the input of this filter. Find the output $y(t)$.

Solution:

The filter has ideal phase response and magnitude response.

$$g(t) \Leftrightarrow G(\omega) ; y(t) \Leftrightarrow Y(\omega) \text{ and}$$

$$Y(\omega) = G(\omega) H(\omega)$$

$$= G(\omega) (1 + K \cos T\omega) e^{-j\omega t_d}$$

$$Y(\omega) = G(\omega) e^{-j\omega t_d} + K [G(\omega) \cos T\omega] e^{-j\omega t_d}$$

Using time shifting property

$$g(t-t_0) \Leftrightarrow G(\omega) e^{-j\omega t_0}$$

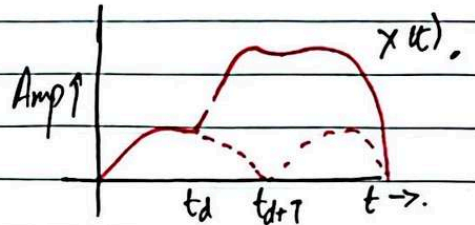
$$g(t-T) + g(t+T) \Leftrightarrow 2G(\omega) \cos T\omega$$

The time domain equivalent of the equation can be expressed as

$$y(t) = g(t-t_d) + \frac{k}{2} [g(t-t_d-T) + g(t-t_d+T)]$$

The output $y(t)$ is actually

$$g(t) + \frac{k}{2} [g(t-T) + g(t+T)] \text{ delayed by}$$



t_d .
 It consists of $g(t)$ and its echoes shifted by $\pm t_d$.

$$g(t-t_0) \Leftrightarrow G(\omega) e^{-j\omega t_0}$$

$$g(t-T) + g(t+T) \Leftrightarrow 2G(\omega) \cos T\omega$$

The time domain equivalent of the equation can be expressed as

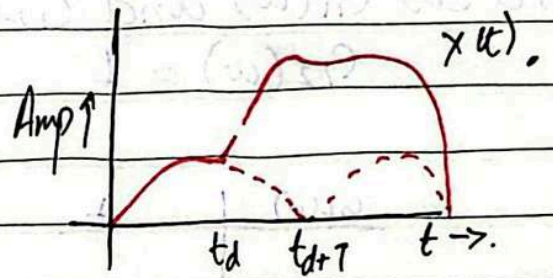
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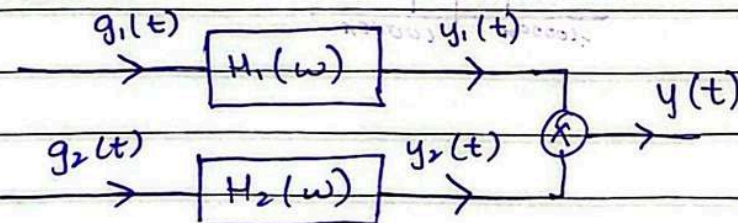
t_d .

It consists of $g(t)$ and its echoes shifted by $\pm t_d$.



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Question :-



Signals $g_1(t) = 10^4 \text{rect}(10^4 t)$ and the $g_2(t) = \delta(t)$ are applied at the input of the ideal LPF $H_1(\omega) = \text{rect}(\frac{\omega}{20000\pi})$ and $H_2(\omega) = \text{rect}(\frac{\omega}{20000\pi})$. The output $y_1(t)$ and $y_2(t)$ of these filters are multiplied to obtain the signal $y(t) = y_1(t) y_2(t)$.

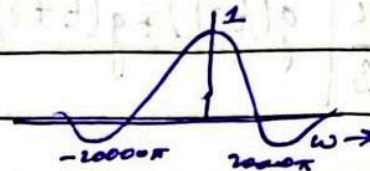
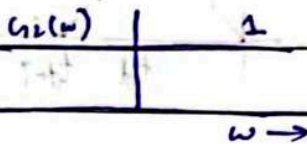
- Sketch $g_1(t)$ and $g_2(t)$
- Sketch $A_1(\omega)$ and $A_2(\omega)$
- Sketch $Y_1(\omega)$ and $Y_2(\omega)$
- Find the BW of $Y_1(t)$, $Y_2(t)$ and $Y(t)$

Solution:-

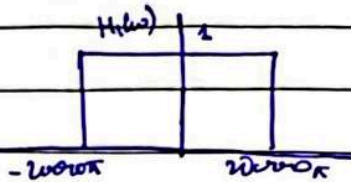
- Sketch $G_1(\omega)$ and $G_2(\omega)$

$$G_2(\omega) = 1$$

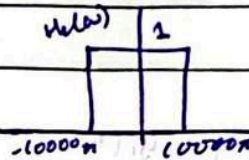
$$G_1(\omega) = \text{sinc}\left(\frac{\omega}{20000}\right)$$



- Sketch $H_1(\omega)$ and $H_2(\omega)$



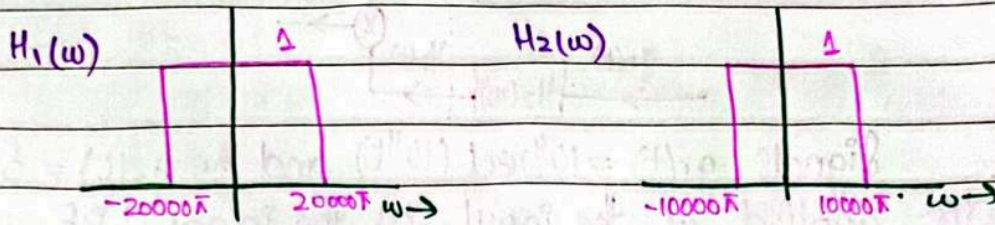
P-VACT



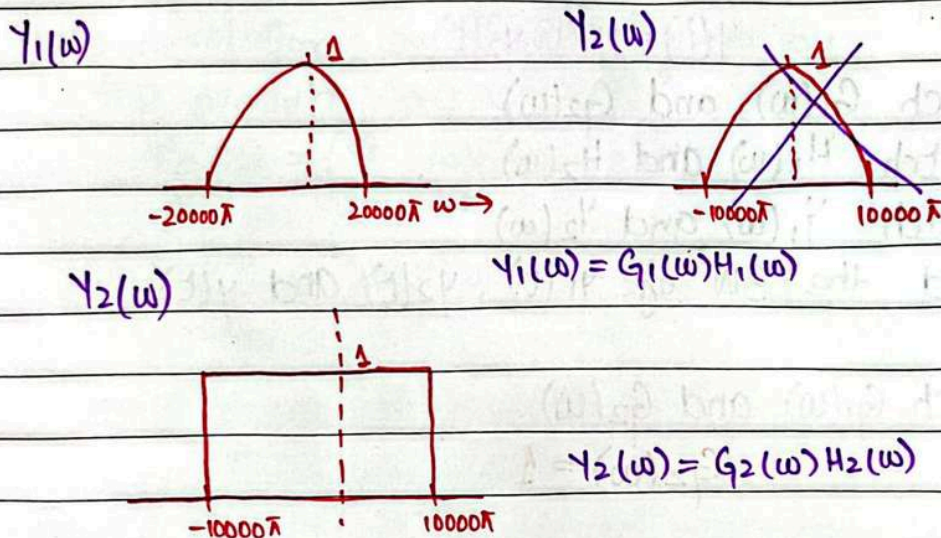
c)



b) Sketch $H_1(\omega)$ and $H_2(\omega)$



c) Sketch $\gamma_1(\omega)$ and $\gamma_2(\omega)$

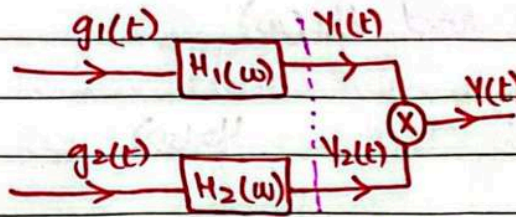


d) Find the BW of $\gamma_1(t)$, $\gamma_2(t)$ and $\gamma(t)$

$$\text{BW of } \gamma_1(\omega) = \frac{20000\pi}{2\pi} = 10000 \text{ Hz} = 10 \text{ kHz}$$

$$\text{BW of } \gamma_2(\omega) = \frac{10000\pi}{2\pi} = 5000 \text{ Hz} = 5 \text{ kHz}$$

$$\text{BW of } \gamma(\omega) = \gamma_1(\omega) + \gamma_2(\omega) = 10 + 5 = 15 \text{ kHz}$$

Q:

Signals $g_1(t) = 10^4 \text{rect}(10^4 t)$ and the $g_2(t) = \delta(t)$ are applied at the input of the ideal LPF

$$H_1(w) = \text{rect}\left(\frac{w}{40000\pi}\right) \text{ and } H_2(w) = \text{rect}\left(\frac{w}{20000\pi}\right)$$

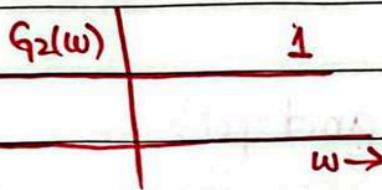
The outputs $y_1(t)$ and $y_2(t)$ of these filters are multiplied to obtain the signal

$$y(t) = y_1(t) y_2(t)$$

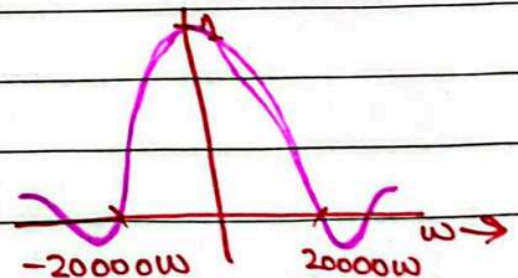
- Sketch $G_1(w)$ and $G_2(w)$
- Sketch $H_1(w)$ and $H_2(w)$
- Sketch $Y_1(w)$ and $Y_2(w)$
- Find the BW of $y_1(t)$, $y_2(t)$ and $y(t)$

a) Sketch $G_1(w)$ and $G_2(w)$

$$G_2(w) = 1$$



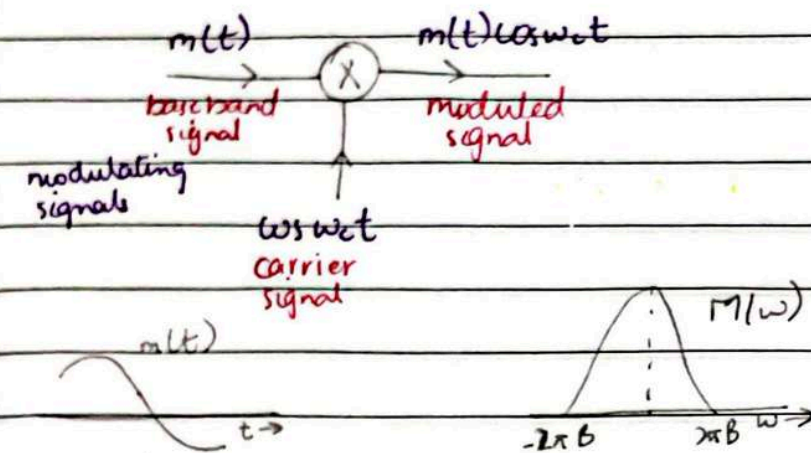
$$G_1(w) = \text{sinc}\left(\frac{w}{20000}\right)$$



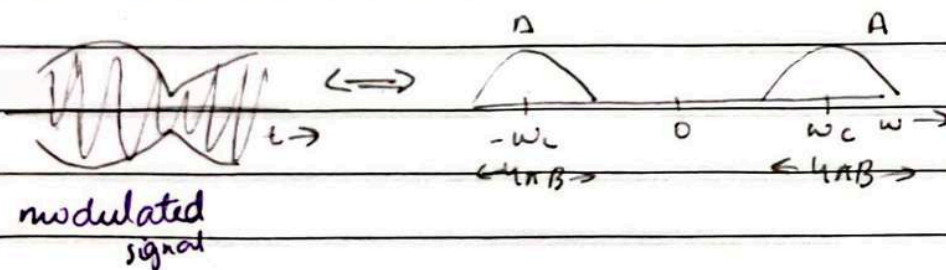
DAY-10

Amplitude Modulation (AM)

modulation



modulating signal



modulated signal

$$m(t) \longleftrightarrow M(\omega)$$

$$m(t) \cos \omega_c t \Leftrightarrow \frac{1}{2} [M(\omega + \omega_c) + M(\omega - \omega_c)]$$

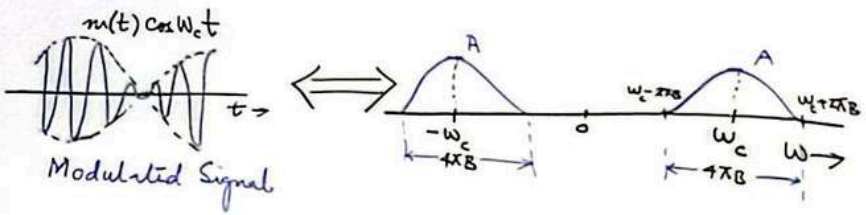
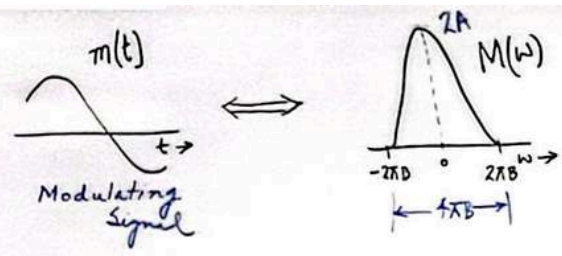
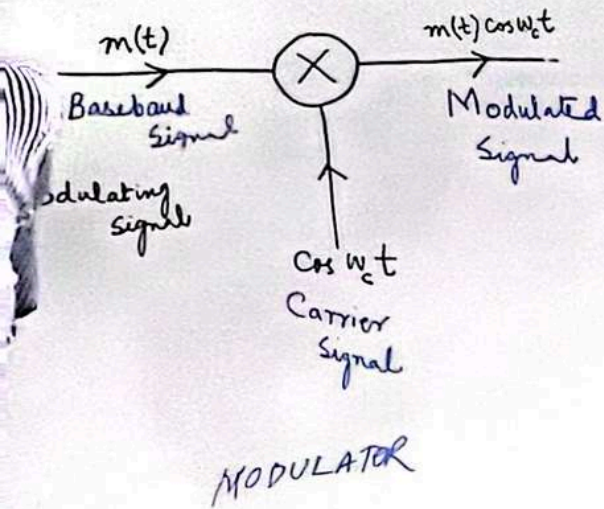
$$\underset{\substack{\downarrow \\ \text{AMP}}}{A} \cos \left(\underset{\substack{\downarrow \\ \text{freq}}}{\omega_c} t + \underset{\substack{\downarrow \\ \text{phase}}}{\theta_c} \right)$$

In AM A would vary in proportion to $m(t)$.

Amplitude modulation is defined as a process in which the amplitude of the carrier wave is varied about a mean value, linearly with the baseband signal $m(t)$.

Amplitude modulation (AM)

Modulation



$$m(t) \longleftrightarrow M(w)$$

$$m(t) \cos w_c t \longleftrightarrow \frac{1}{2} [M(w + w_c) + M(w - w_c)]$$

Amplitude modulation is defined as a process in which the amplitude of the carrier wave is varied about a mean value, linearly with the baseband signal $m(t)$.

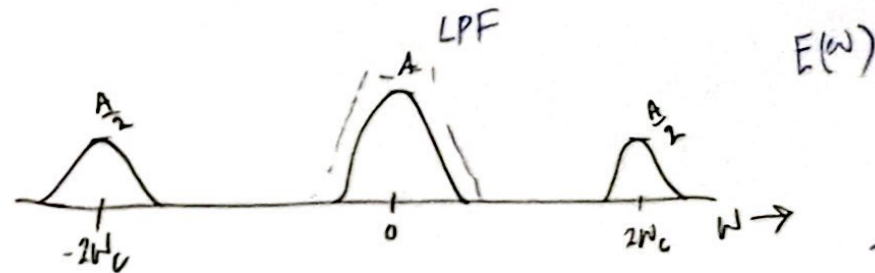
$$A \cos(\omega_c t + \theta_c)$$

$\downarrow$ Amp $\downarrow$ Freq $\downarrow$ Phase

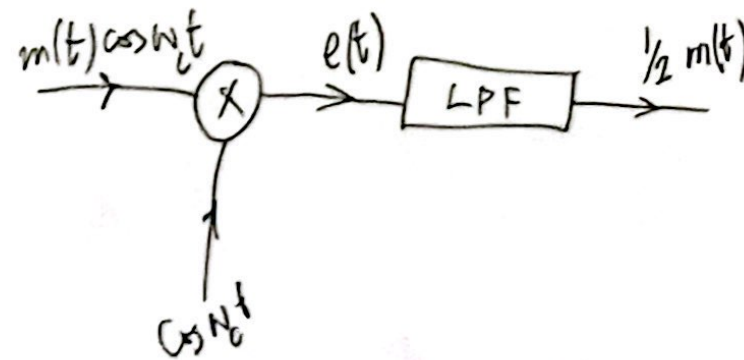
In AM, A vary in proportion to m(t).

Demodulation consists of multiplication of incoming modulated signal $m(t)\cos\omega_c t$ by the carrier $\cos\omega_c t$

$$\begin{aligned}
 e(t) &= m(t)\cos^2\omega_c t \\
 &= \frac{1}{2} [m(t) + m(t)\cos 2\omega_c t] \\
 E(\omega) &= \frac{1}{2} M(\omega) + \frac{1}{4} [M(\omega + 2\omega_c) + M(\omega - 2\omega_c)]
 \end{aligned}$$



Demodulator



$$m(t) \cos \omega_c t \Leftrightarrow \frac{1}{2} [M(\omega + \omega_c) + M(\omega - \omega_c)]$$

$$A \cos(\omega_c t + \theta_c)$$

↓ ↓ ↓
 Amp freq phase

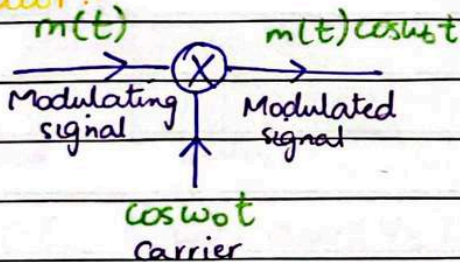
In AM, A would vary in proportion to $m(t)$.

Amplitude modulation is defined as a process in which the amplitude of the carrier wave is varied about a mean value, linearly with the baseband signal $m(t)$.

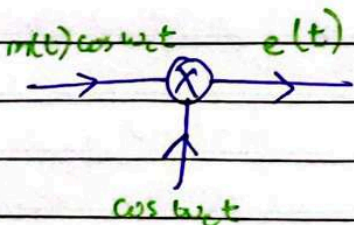
(AFTER MID 1) DAY-11

Amplitude Modulation :-

Modulator :-



Demodulator :-



Synchronous:

$$\begin{aligned} e(t) &= m(t) \cos^2 \omega_c t \\ &= \frac{1}{2} [m(t) + m(t) \cos 2\omega_c t] \\ &= \frac{1}{2} m(t) + \frac{1}{2} m(t) \cos 2\omega_c t \end{aligned}$$

Types Of Modulator :-

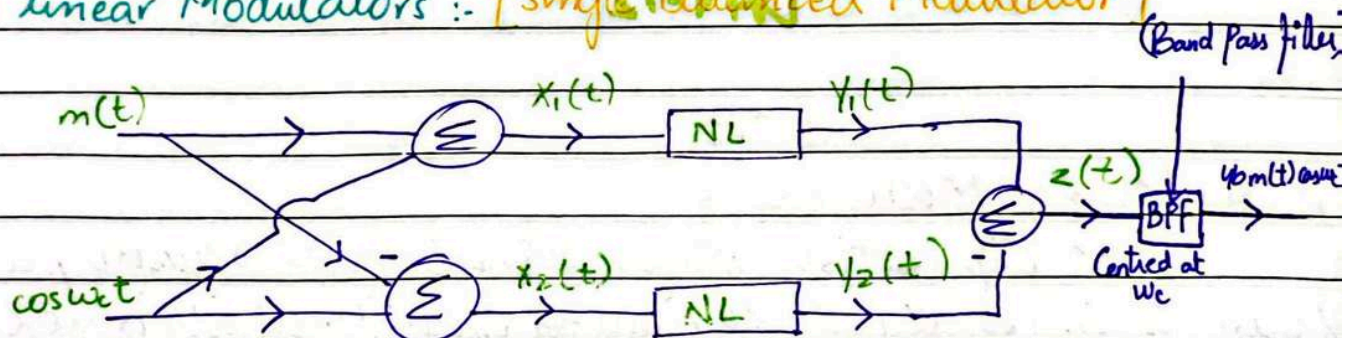
1. Multiplier Modulator :-

Analog multiplier with o/p proportional to the two

i/p s -

→ Difficult to maintain linearity

Non-Linear Modulators :- [Single Balanced Modulator]



$$y(t) = ax(t) + bx^2(t).$$

$$z(t) = y_1(t) - y_2(t) = [ax_1(t) + bx_1^2(t)] - [ax_2(t) + bx_2^2(t)]$$

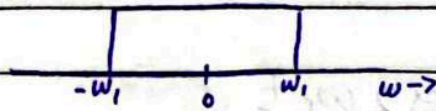
Put $x_1(t) = \cos \omega_c t + m(t).$

$x_2(t) = \cos \omega_c t - m(t).$

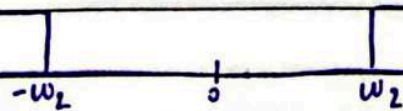
$$z(t) = 2am(t) + 4bm(t) \cos \omega_c t.$$

Explanation with Example why we won't use LPF, HPF.

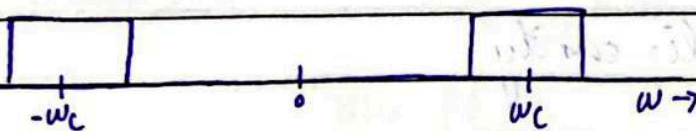
LPF



HPF



BPF

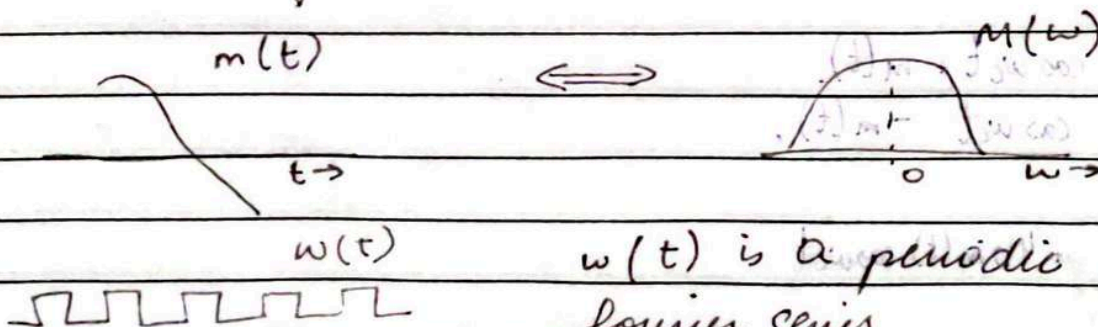


DAY-12

Amplitude Modulators

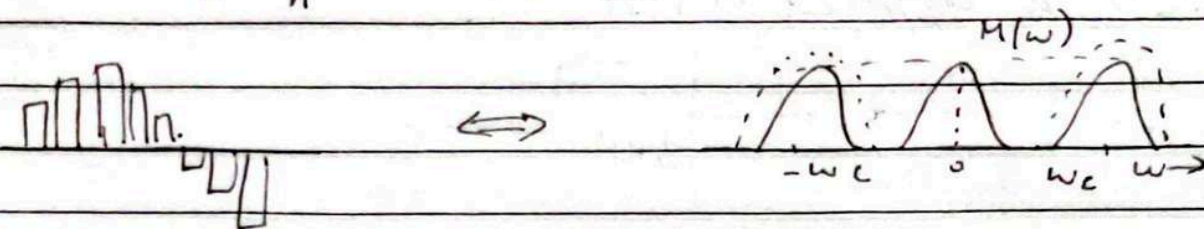
1) Switching Modulators:

Multiplication of a signal by a square pulse train is in reality a switching operation which involves switching the signal $m(t)$ on and off periodically and can be accomplished by simple switching elements controlled by a pulse train $w(t)$.

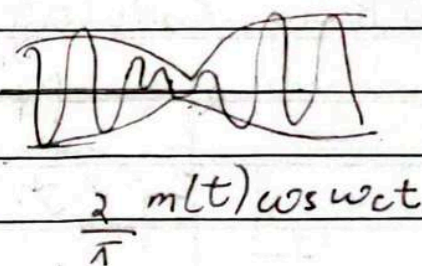
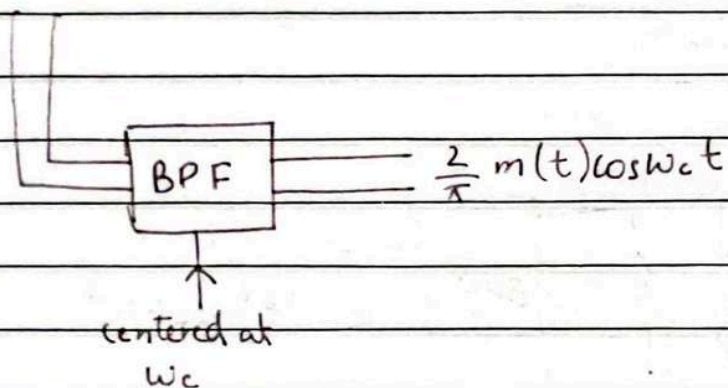


400

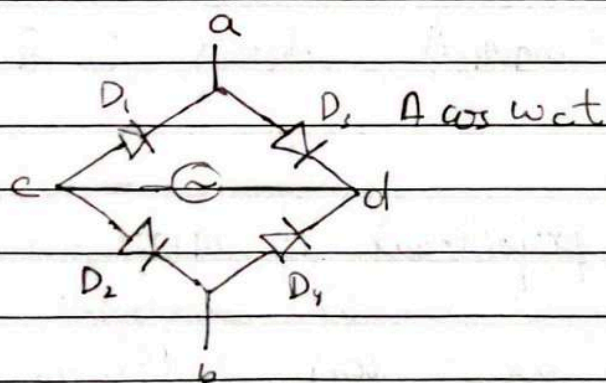
$$w(t) = \frac{1}{2} + \frac{2}{\pi} (\cos w_c t - \frac{1}{3} \cos 3w_c t + \frac{1}{5} \cos 5w_c t - \dots)$$



$$m(t)w(t) = \frac{1}{2} m(t) + \frac{2}{\pi} (m(t) \cos w_c t - \frac{1}{3} m(t) \cos 3w_c t + \dots)$$

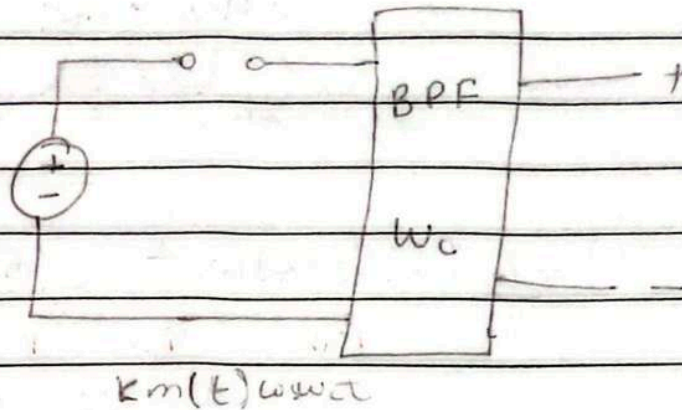


The switching operation can be achieved with a diode - bridge modulator (which is a double balanced modulator)

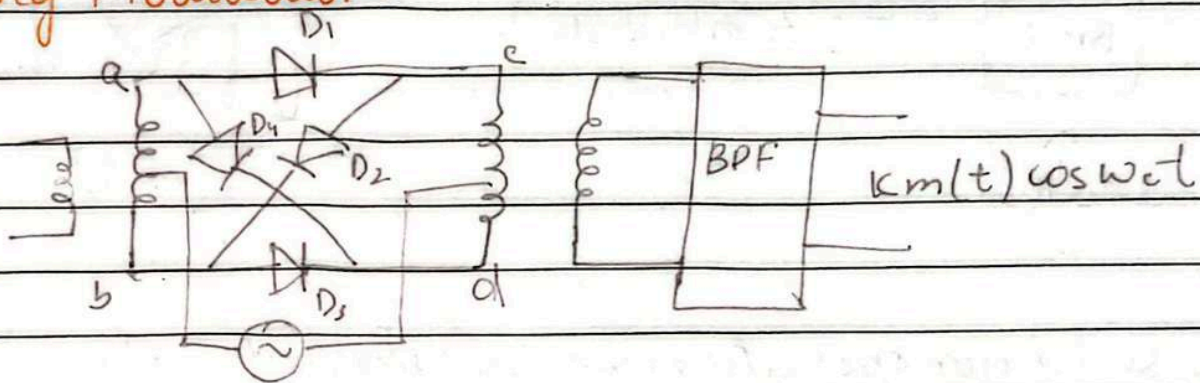


Diode bridge serves as the desired electronic switch when terminals a and b open and close

periodically with carrier frequency f_c when $A \cos \omega_c t$ is applied across cd.



2) Ring Modulator :-



During +ve half cycle of the carrier :-

Diode D_1 & D_3 conducts whereas D_2 & D_4 are open

⇒ o/p proportional to $m(t)$

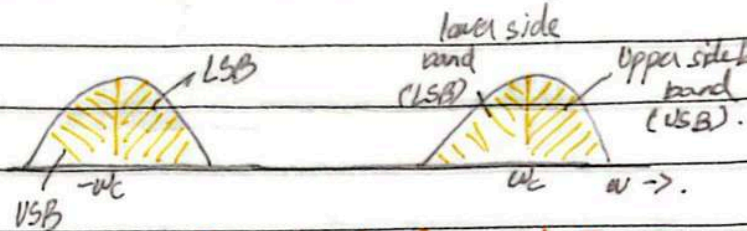
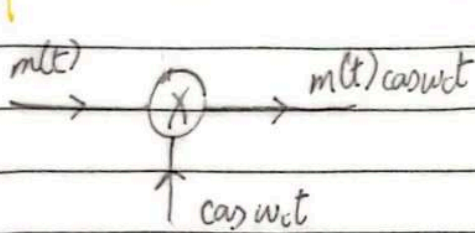
During -ve half cycle of the carrier :-

Diode D_1 & D_3 are open whereas D_2 & D_4 conducts

⇒ o/p proportional to $-m(t)$

Day - 13

Amplitude Modulation: (With Carrier - WC)



Double Side band - Suppressed Carrier.
(DSB - SC).

For AM $\begin{cases} \text{SC (costly sophisticated receiver)} \\ \text{WC.} \end{cases}$ [Synchronous].
 [Point-to-multipoint system]

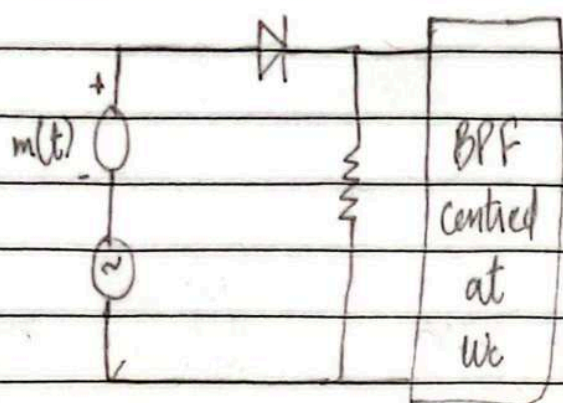
$$\begin{aligned} \phi_{AM}(t) &= A \cos w_c t + m(t) \cos w_c t \\ &= [A + m(t)] \cos w_c t. \end{aligned}$$

$$\phi_{AM}(w) = \frac{1}{2} [M(w - w_c) + M(w + w_c)] + \pi A [\delta(w + w_c) + \delta(w - w_c)].$$

Two additional pulses at w_c & $-w_c$

Generation of AM - WC Signal:-

AM signals can be generated by DSB-SC modulator if modulating signal is $[A + m(t)]$.



ci-180

Input is $\cos \omega_c t + m(t)$ with $c \gg m(t)$. Diode act as a switching element and multiplies signal $[c \cos \omega_c t + m(t)]$ pulse train $\omega(t)$.

$$\begin{aligned}
 V_{in}(t) &= [c \cos \omega_c t + m(t)] \cdot \omega(t) \\
 &= [c \cos \omega_c t + m(t)] \cdot \left\{ \frac{1}{2} + \frac{2}{\pi} \left(\cos \omega_c t - \frac{1}{3} \cos 3\omega_c t + \frac{1}{5} \cos 5\omega_c t - \dots \right) \right\} \\
 &= \frac{c}{2} \cos \omega_c t + \frac{2}{\pi} m(t) \cos \omega_c t + \dots
 \end{aligned}$$

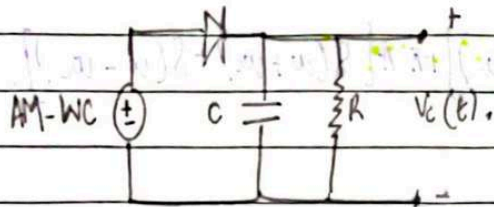
Suppressed by BPF (ω_c)

$$V_{cc}(t) = \underbrace{\frac{c}{2} \cos \omega_c t}_{\text{carrier}} + \underbrace{\frac{2}{\pi} m(t) \cos \omega_c t}_{\text{modulated signal}}$$

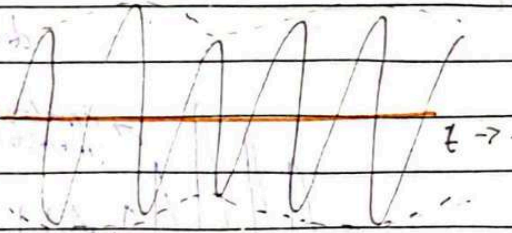
AM-WC.

Envelope Detector:

In an envelope detector, the output of detector follows envelope of modulated signal.



On the positive cycle, the diode conducts and capacitor charges upto peak voltage. During the negative cycle the diode is open & capacitor discharges through the resistor R at a slow rate. Depending on the time constant RC the output $V_C(t)$ closely follows envelope of input.



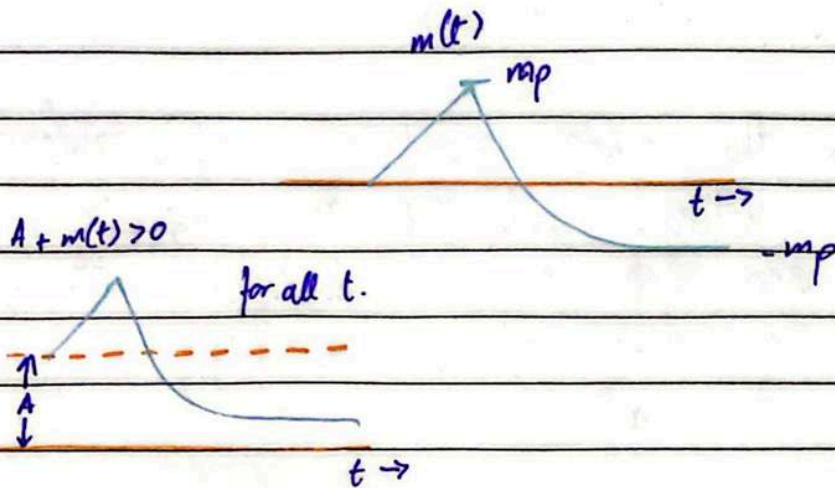
DAY-14.

Amplitude Modulation for BroadCast System as opposed to point to point communication.

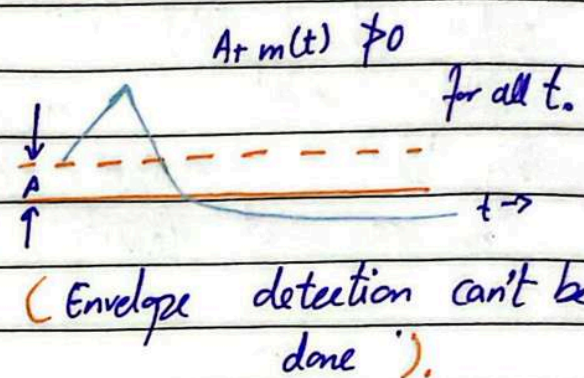
$$\begin{aligned}\phi_{AM}(t) &= A \cos \omega_c t + m(t) \cos \omega_c t \\ &= [A + m(t)] \cos \omega_c t.\end{aligned}$$

The spectrum of $\phi_{AM}(t)$ is same as that of $m(t) \cos \omega_c t$ plus two additional impulses at ω_c .

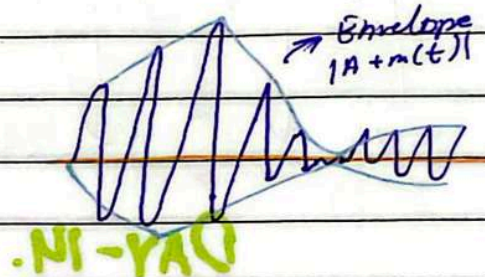
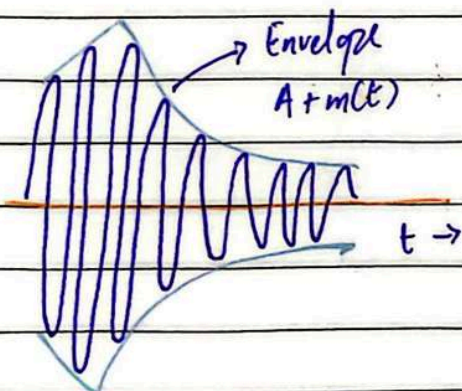
$$\phi_{AM}(t) \leftrightarrow \frac{1}{2} [M(\omega + \omega_c) + M(\omega - \omega_c)] + \pi A [\delta(\omega + \omega_c) + \delta(\omega - \omega_c)]$$



(Envelope detection can be done.)



(Envelope detection can't be done).



$A + m(t) > 0$ for all t
 Modulation index $\mu = \frac{m_p}{A}$

$A + m(t) \geq 0$ implies that
 $A \geq m_p$

envelope detection to be possible

$0 \leq \mu \leq 1$

$\mu > 1$ is "overmodulation" $\rightarrow$ envelope detection not possible

Conditions For envelope Detection:-

$$A + m(t) \geq 0 \text{ for all } t$$

$$m(t) \geq -m_p$$

$$A - m_p \geq 0$$

$$A \geq m_p$$

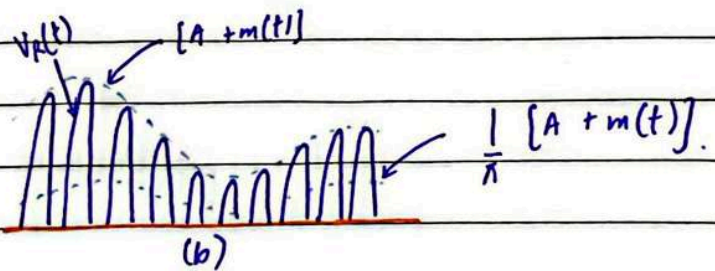
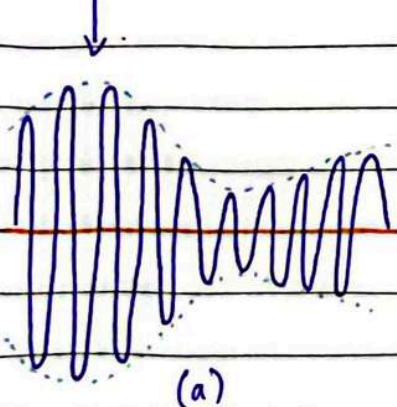
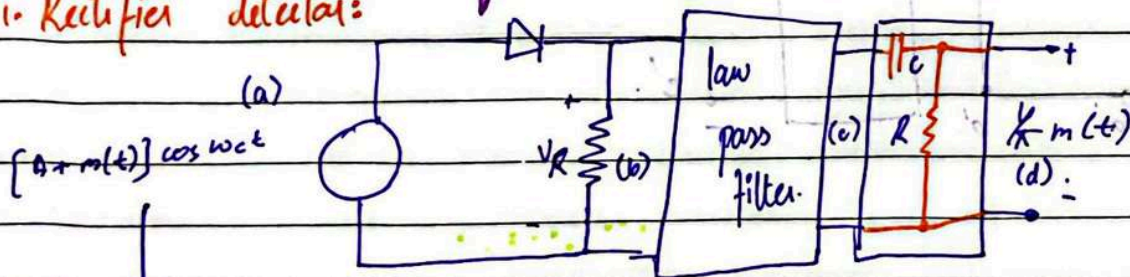
$$\mu = \frac{m_p}{A}$$

(Modulation Index)

$$\text{Range } 0 \leq \mu \leq 1$$

Demodulation of AM signals:-

1. Rectifier detector:



$$\frac{1}{\pi} [A + m(t)]$$

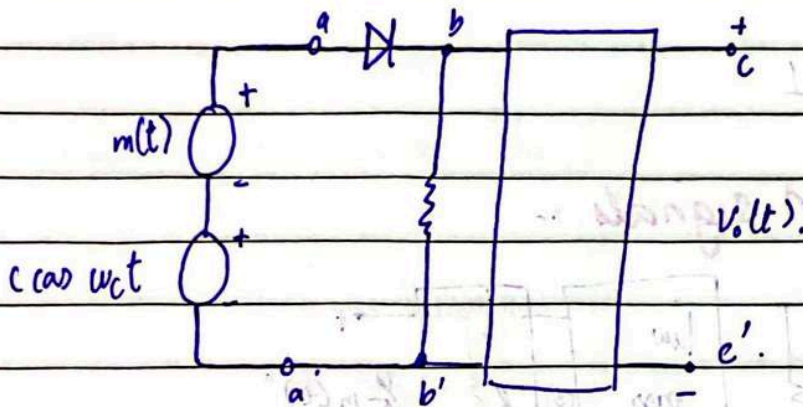
(c)

$$\frac{1}{\pi} m(t)$$

(d)

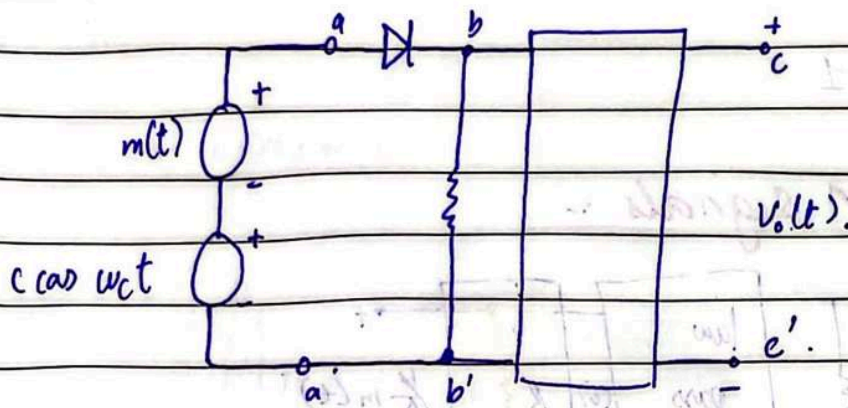
Generation of AM Signals.

- AM Signals can be generated using any of DSB-SC modulators if modulating signal is $A + m(t)$ instead of just $m(t)$.
- No need to have a balanced circuit, accounts to a simple modulators for AM.



Generation of AM Signals.

- AM Signals can be generated using any of DSB-SC modulators if modulating signal is $A + m(t)$ instead of just $m(t)$.
- No need to have a balanced circuit, accounts to a simple modulators for AM.



$x * x = V$ $x * x = V$
DAY-15
 Consecutively
DAY-15

Modulation Index

$$\mu = \frac{m_p}{A}$$

Example:

Sketch $\phi(t)$ for modulation indices of $\mu = 0.5$ and $\mu = 1$, when $m(t) = B \cos \omega_m t$. This case is referred to as

Tone modulation because the modulating signal is a pure sinusoid (or tone)

Tone modulation :- A modulation in which message and carrier signal both are sinusoid.

sol.

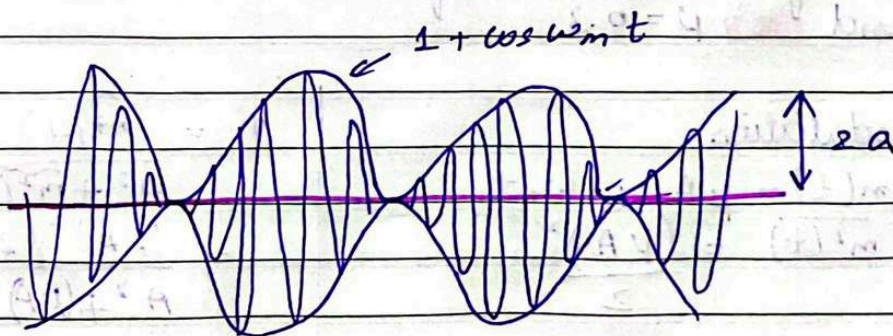
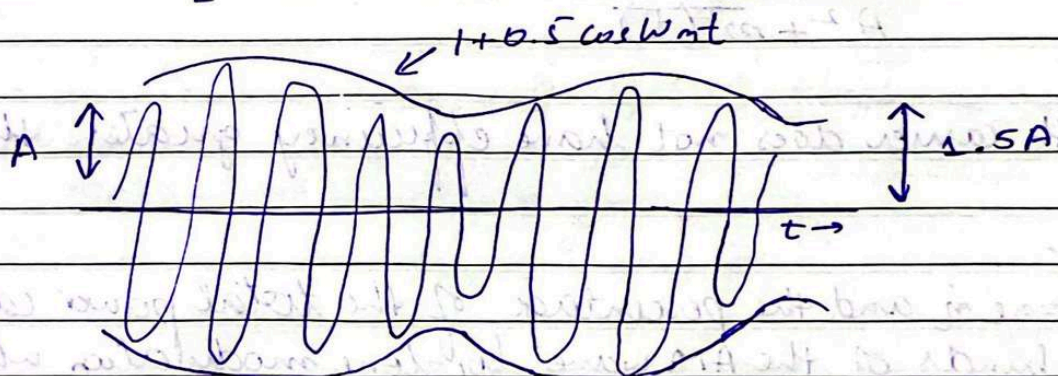
$$\mu = \frac{B}{A}$$

Hence $B = \mu A$ and

$$m(t) = B \cos \omega_m t = \mu A \cos \omega_m t$$

therefore

$$\phi_{AM}(t) = [A + m(t)] \cos \omega_c t = A [1 + \mu \cos \omega_m t] \cos \omega_c t$$



Sideband And Carrier Power:

$$\phi_{AM}(t) = \underbrace{A \cos \omega_c t}_{\text{carrier}} + \underbrace{m(t) \cos \omega_c t}_{\text{sideband}}$$

carrier power

$$P_c = \frac{A^2}{2}$$

sideband power

$$P_s = \frac{1}{2} \overline{m^2(t)}$$

$$\eta = \frac{\text{useful power}}{\text{total power}}$$

$$= \frac{P_s}{P_s + P_c}$$

$$\eta = \frac{\overline{m^2(t)}}{A^2 + \overline{m^2(t)}} \times 100\%$$

AM with carrier does not have efficiency greater than 25.

Example:

Determine η and the percentage of the total power carried by the sidebands of the AM wave for tone modulation when $\mu = 0.5$ and $\mu = 0.3$

sol

For tone modulation

$$m(t) = \mu A \cos \omega_m t$$

$$\overline{m^2(t)} = \frac{(\mu A)^2}{2}$$

$$\begin{aligned} \eta &= \frac{\overline{m^2(t)}}{A^2 + \overline{m^2(t)}} \\ &= \frac{(\mu A)^2 / 2}{A^2 + (\mu A)^2 / 2} \end{aligned}$$

For $\mu = 0.5$

$$\eta = \frac{\mu^2}{2 + \mu^2} \times 100\%$$

$$= \frac{(0.5)^2}{2 + (0.5)^2} \times 100\%$$

$$\eta = 11.11\%$$

$$= \frac{\mu^2}{2 + \mu^2}$$

For $\mu = 0.3$

$$\eta = \frac{(0.3)^2}{2 + (0.3)^2} \times 100\%$$

$$\eta = 4.3\%$$

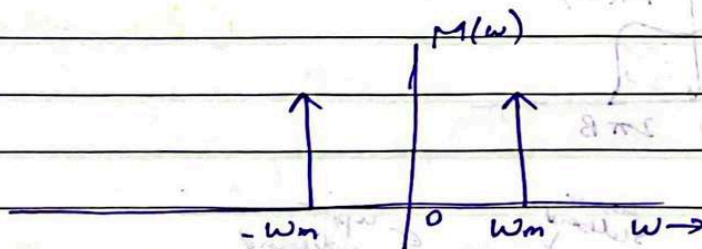
Tone Modulation :-

Example:-

Find the DSB-SC signal and sketch its spectrum. Identify USB and LSB.

$$m(t) = \cos \omega_m t$$

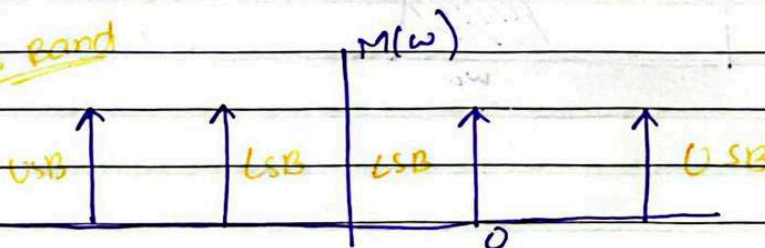
$$M(\omega) = \pi [\delta(\omega - \omega_m) + \delta(\omega + \omega_m)]$$

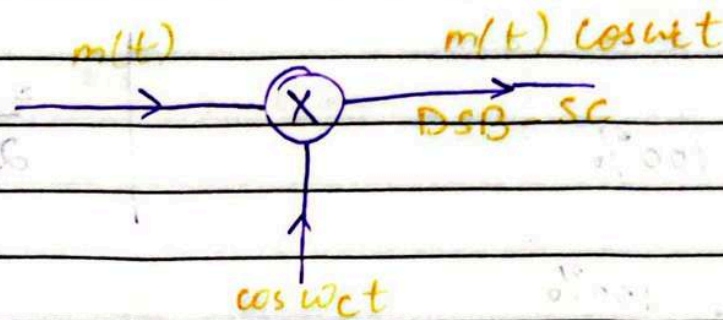


(or)

$$\omega = 2\pi f$$

DSB Band





For the baseband signal

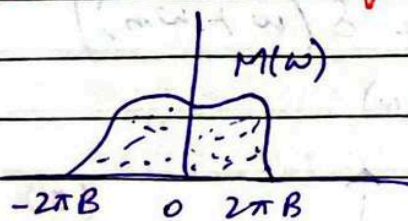
$$m(t) = \cos w_m t$$

the DSB-SC signal $\phi_{DSB-SC}(t)$ is

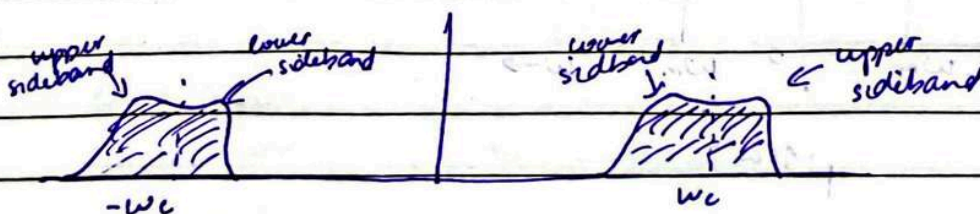
$$\begin{aligned}\phi_{DSB-SC}(t) &= m(t) \cos w_c t \\ &= \cos w_m t \cos w_c t \\ &= \frac{1}{2} [\cos (w_c + w_m)t + \cos (w_c - w_m)t]\end{aligned}$$

DAV-16

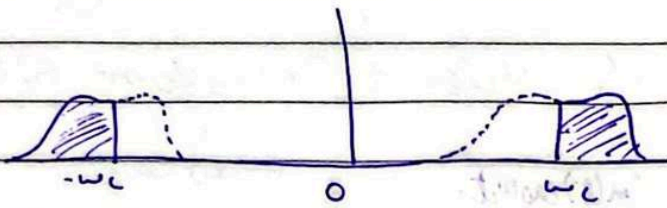
Amplitude Modulation : Single Side-Band (SSB)



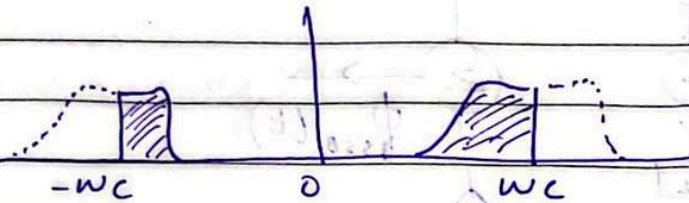
a) Baseband



b) DSB



c) USB



d) LSB

SSB signals without any additional carrier, hence, they are suppressed carrier signals (SSB-SC)

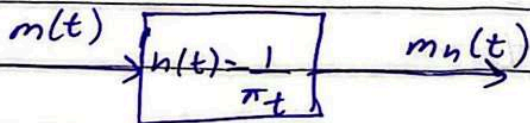
Hilbert Transform :-

$m(t)$ $m_h(t)$

$$m_h(t) = m(t) * \frac{1}{\pi t}$$

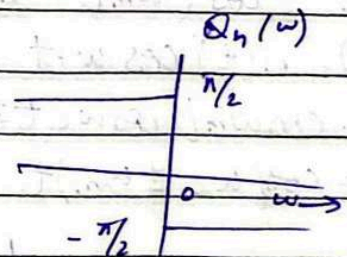
$$m_h(t) = \frac{1}{\pi} \int_{-\infty}^{\infty} \frac{m(\alpha)}{t-\alpha} d\alpha$$

$m_h(t)$ is hilbert transform of $m(t)$

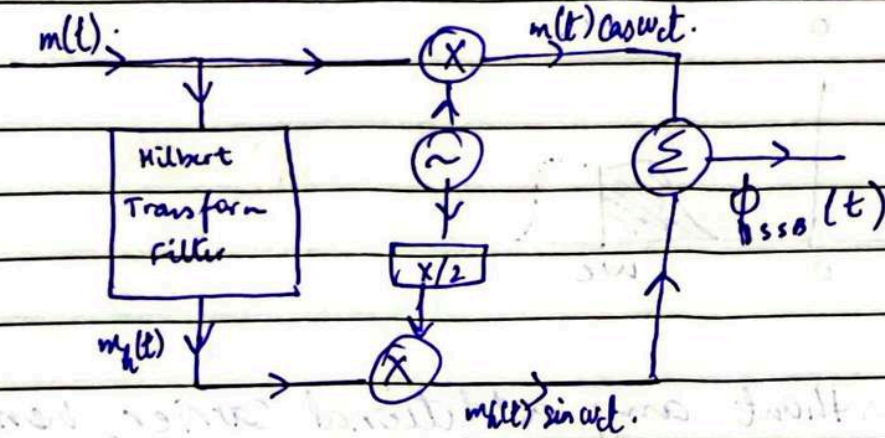


$$H(\omega) = -j \operatorname{sgn}(\omega)$$

$$= \begin{cases} -j = e^{-j\pi/2} & \omega > 0 \\ j = e^{j\pi/2} & \omega < 0 \end{cases}$$



Generation of SSB using the Hilbert Transform Filter ::



$$\phi_{LSB}(t) = m(t) \cos wct + m_h(t) \sin wct.$$

$$\phi_{USB}(t) = m(t) \cos wct - m_h(t) \sin wct.$$

Example 4.7

Find $\phi(t)$ when modulating signal is a sinusoid

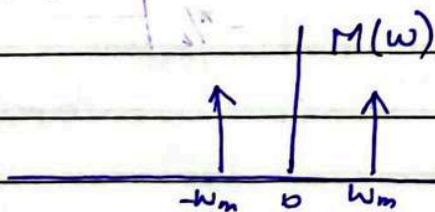
$$m(t) = \cos w_m t.$$

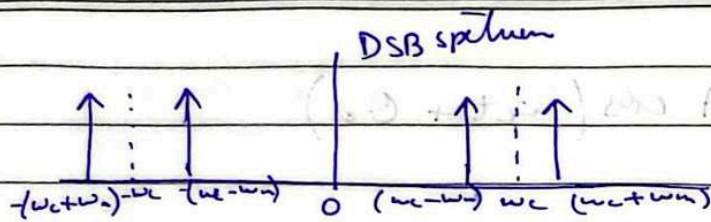
Sol

$$m_h(t) = \cos(w_m t - \pi/2) = \sin w_m t$$

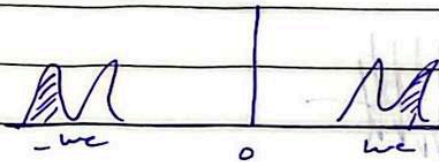
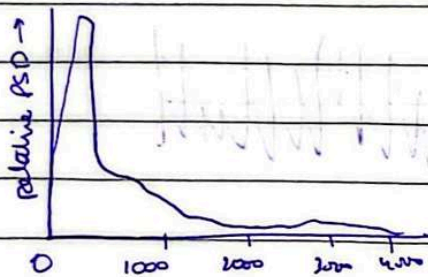
$$\phi_{ssb}(t) = m(t) \cos w_c t \mp m_h(t) \sin w_c t$$

$$\begin{aligned} \phi_{ssb}(t) &= \cos w_m t \cos w_c t \mp \sin w_m t \sin w_c t \\ &= \cos(w_c \pm w_m)t \end{aligned}$$





Selective-Filtering Method



DAY-17 (Chapter 5)

Angle Modulation ::

Techniques of modulation where the angle of the carrier is varied with a modulating signal $m(t)$ are known as angle modulation.

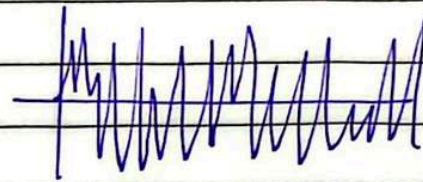
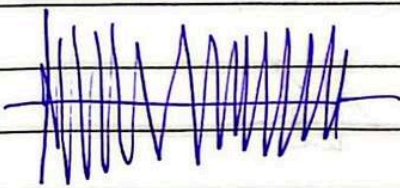
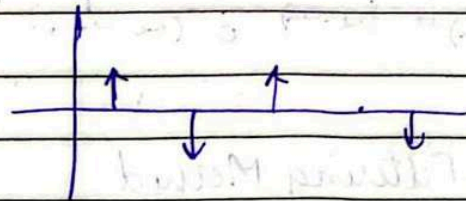
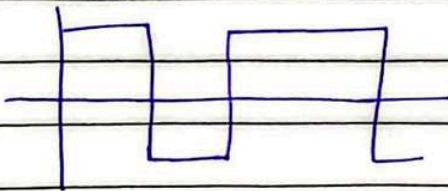
Two possibilities are phase modulation (PM) and frequency modulation (FM).

$$\phi_{PM}(t) = A \cos [w_c t + k_p m(t)]$$

$$\phi_{FM}(t) = A \cos [w_c t + k_f \int_{-\infty}^t m(\alpha) d\alpha]$$

w_i = instantaneous frequency

Generalized sinusoid $A \cos(\omega_c t + \theta_0)$



Instantaneous Frequency :-

$$\phi(t) = A \cos(\omega_c t + \theta_0)$$

The instantaneous frequency ω_i at any instant t is the slope of $\theta(t)$ at t . This for $\phi(t)$

$$\omega_i(t) = \frac{d\theta}{dt}$$

$$\theta(t) = \int_{-\infty}^t \omega_i(\alpha) d\alpha$$

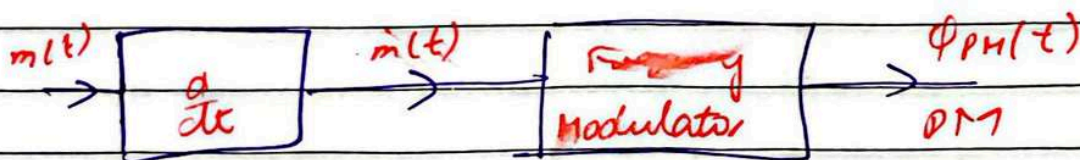
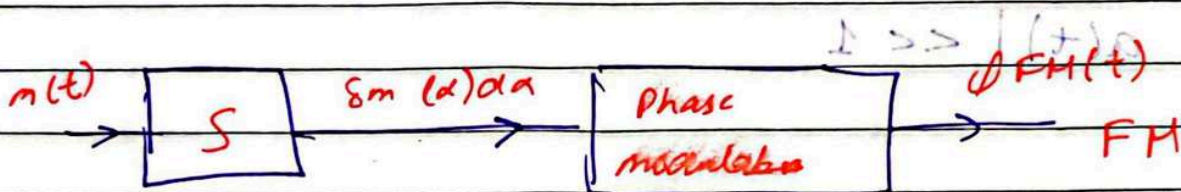
the possibility of losing the information of $m(t)$ by varying angle θ of the carrier arises

$$\phi_{PM}(t) = A \cos[\omega_c t + k_p m(t)]$$

$$\phi_{FM}(t) = A \cos[\omega_c t + k_f \int_0^t m(\alpha) d\alpha]$$

Instantaneous frequency varies linearly with modulating signal in FM.

Generalized Concept of Modulation: $A = (1) \dots$



Bandwidth of angle modulator carrier..

Generalized angle-modulator carrier

$$\begin{aligned}\phi_{FH}(t) &= A \cos[\omega_c t + \psi(t)] \\ &= A \cos\left[\omega_c t + \int_{-\infty}^t m(\alpha) h(t-\alpha) d\alpha\right]\end{aligned}$$

If $h(t) = k_p \delta(t) \dots \dots \dots$ PM

$h(t) = k_f u(t) \dots \dots \dots$ FM

Let us define $q(t) = \int_{-\infty}^t m(\alpha) d\alpha$

$$\begin{aligned}\phi_{FH}(t) &= A e^{j[\omega_c t + k_f q(t)]} \\ \text{Shog} &= A e^{j k_f q(t)} e^{j \omega_c t}\end{aligned}$$

$$\begin{aligned}\phi_{FH}(t) &= \text{Re } \phi_{FH}(t) = A [\cos \omega_c t - k_f q(t) \sin \omega_c t \\ &\quad - \frac{k_f^2}{2!} q^2(t) \cos \omega_c t + \frac{k_f^3}{3!} q^3(t) \sin \omega_c t \dots]\end{aligned}$$

Theoretical BW of an FM wave is infinite

NB FM

$$\phi_{FM}(t) \approx A [\cos \omega_c t - k_f a(t) \sin \omega_c t]$$

given that

$$|k_f a(t)| \ll 1$$

Theoretical BW of an FM wave is infinite

NB FM

$$\phi_{FM}(t) \approx A [\cos \omega_c t - k_f a(t) \sin \omega_c t]$$

given that

$$|k_f a(t)| \ll 1$$

DAY-18

Bandwidth of FM waves :-

Theoretically an FM modulated wave is not bandlimited.

NB FM For very small k_f (narrowband FM)

$$\phi_{FM}(t) = A [\cos \omega_c t - k_f a(t) \sin \omega_c t]$$

given that $|k_f a(t)| \ll 1$

This is linear modulation. The BW of $a(t)$ is B , then the BW of $\phi_{FM}(t)$ would only be $2B$.

Approximate BW for WB FM (Wide Band FM)

$$B_{FM} = \frac{1}{2\pi} (2k_f m_p + 8\pi B)$$

$$B_{FM} \approx 2(\Delta f + B) \dots (1)$$

For a truly wideband case, where $\Delta f \gg B$, FM BW can be approximated as

$$B_{FM} \approx 2\Delta f \quad \Delta f \gg B \dots (2)$$

We define the deviation ratio $\beta = \frac{\Delta f}{f_m}$

$$B_{FM} = 2B(\beta + 1) \dots \dots (3)$$

Example:

$$\phi_m(t) = 10 \cos(\omega_c t + 5 \sin 3000t + 10 \sin 2000\pi t)$$

- power of angle modulated wave
- frequency deviation Δf
- Deviation ratio β
- max deviation $\Delta \phi$
- Band width of $\phi_m(t)$

Sol

$$a) \frac{A^2}{2} = \frac{10^2}{2} = 50$$

- Find instantaneous frequency ω_i

$$\omega_i = \frac{d}{dt} \theta(t) = \omega_c + 15000 \cos 3000t + 20000\pi \cos 2000\pi t$$

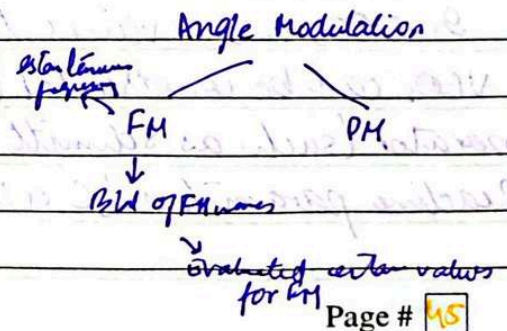
$$\text{carrier deviation } 15000 \cos 3000t + 20000\pi \cos 2000\pi t$$

$$\Delta \omega = 15000 + 20000\pi \text{ (add in phase)}$$

DAY-19

FM Modulation and Demodulation :-

Frequency modulators are
time varying non-linear systems



Oscillators (voltage controlled oscillator VCO) are designed to produce frequency modulators.

Oscillator design $\begin{cases} \text{Reactance Tube} \\ \text{or} \\ \text{Varactor Diode} \end{cases}$

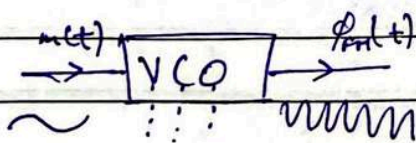
↓
capacitor whose capacitance changes with applied voltage

Direct FM modulation: VCO

Voltage controlled oscillator (VCO) are used to vary the frequency of the carrier signal in accordance with the baseband signal amplitude variations.

Hartley or Colpitt Oscillator:

Frequency of oscillator $\omega_0 = \frac{1}{\sqrt{LC}}$

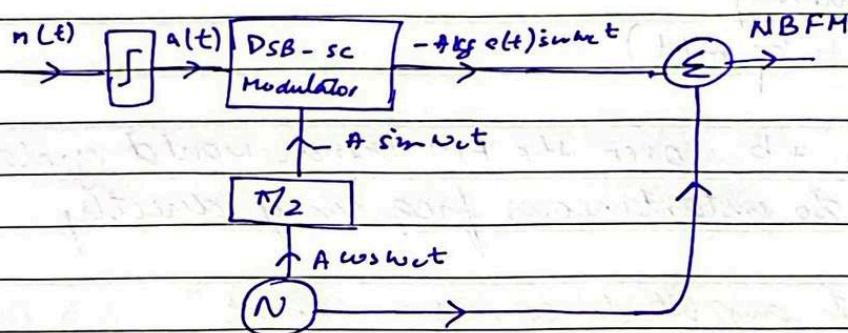


PI-VAD

- In VCOs, ω_0 varies linearly with the controlled voltage.
- VCOs can be constructed using an OR AMP and a hysteresis comparator (such as Schmitt Trigger).
- Reactive parameters (C or L) of the resonant oscillator can be

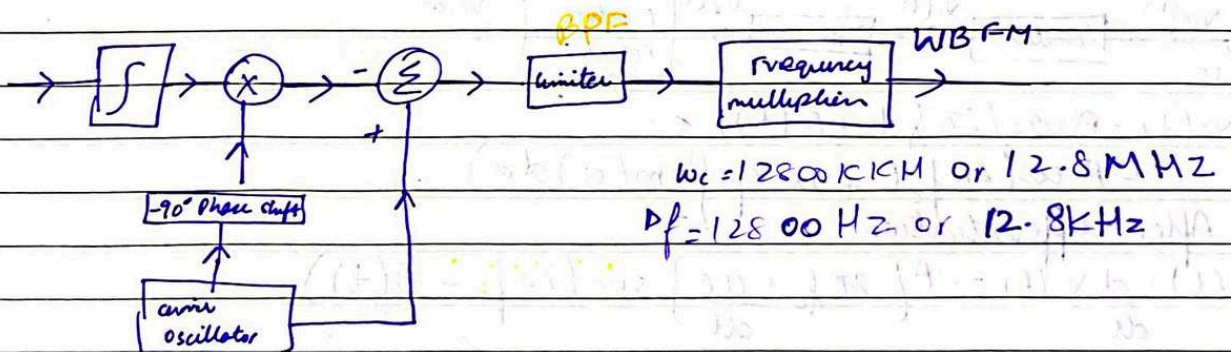
varied to achieve VCO functionality

Narrowband FM wave generation:-



$$\phi_{\text{NBFM}}(t) \approx A [\cos \omega_c t - k_f a(t) \sin \omega_c t]$$

ARMSTRONG'S INDIRECT method of FM Generation:-



A wideband FM signal is produced by multiplying a frequency, the narrowband FM using freq. multiplier

DAY-20

Demodulation of FM :

Information in an FM signal resides in the instantaneous frequency.

$$\omega_i = \omega_c + k_f m(t)$$

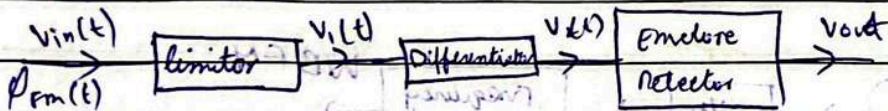
So,

$|H(\omega)| = a\omega + b$ over the FM band, would yield output proportional to instantaneous frequency directly requiring a

"Frequency to amplitude conversion circuit"

1. Slope Detector = Mathematical analysis
2. Zero crossing Detector : Graphical analysis
3. Phase locked loop (PLL)

Slope Detector :



$$\begin{aligned} \rightarrow V_1(t) &= A \cos(2\pi f_c t + \theta(t)) \leftarrow \\ &= A \cos(2\pi f_c t + 2\pi k_f \int_{-\infty}^t m(\alpha) d\alpha) \end{aligned}$$

After differentiation

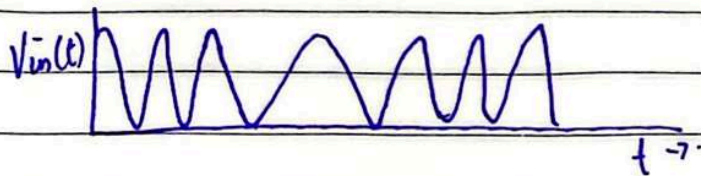
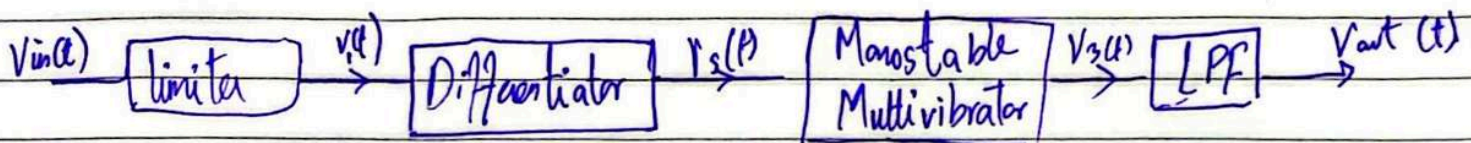
$$V_2(t) = \frac{d}{dt} V_1(t) = -A \left[2\pi f_c + \frac{d\theta}{dt} \right] \sin(2\pi f_c t + \theta(t))$$

FM demodulation can be performed by taking the time derivative often called slope detection of the FM signal, followed by the envelope detection.

$$V_{out}(t) = A \left[2\pi f_c + \frac{d\theta}{dt} \right]$$

$$V_{out}(t) = A 2\pi f_c + A 2\pi k_f \underline{m(t)}$$

Zero Crossing Detector :



∴ frequency to Amplitude conversion.

